Ahmedabad University
School of Engineering and Applied Science

Brain Tumor Segmentation and Localization
Using Traditional Image Processing and OpenCV

Weekly Status Report
Week 7 of 10

Course: ECE501 – Digital Image Processing — **Instructor:** Professor Mehul Raval — **Section:** 01
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1 Introduction

Segmentation of brain tumors is one of the crucial tasks in medical image analysis for the identification of tumors in the brain. MR image localizing and approximate boundaries of abnormal tissue. In this project, traditional image-processing techniques are being considered for brain tumor segmentation without using deep learning. The BraTS 2020 dataset contains 4 MR modalities for every patient: T1, T1ce, T2 and FLAIR. In addition to expert-annotated tumor masks.

The working principle and previous applications of watershed segmentation were studied in Week 6. The A literature review revealed that the use of the traditional watershed transform to an MRI image can Over-segmentation is possible, due to noise and small intensity variations that cause multiple local minima. Marker- to control this process, the controlled watershed can be used to set up a foreground and background area prior to segmentation.

The goal of week 7 is an implementation preview of an implementation with a marker-controlled watershed pipeline with OpenCV. The proposed pipeline is designed to run from the preprocessing to the marker generation, candidate-region extraction and water- shed stages. The output should be a visual preview of the segmentation that will be examined prior to the output: Further refinement and quantitative evaluation was necessary.

2 Objective

This week's focus will be setting up the implementation workflow for watershed based brain tumor.

The specific objectives are:

- Create an MRI slice for watershed processing.
- Apply preprocessing to decrease noise and enhance the quality of image.
- Use intensity based thresholding to localize possible tumor areas.
- Use morphological operations to improve the candidate regions.
- Create foreground, background and unknown-region markers.
- Use OpenCV to apply marker controlled watershed.
- Reflect on the outcome of the region boundaries and areas to improve.

3 Literature Review

The suggested solution to the problem proposed for the implementation is based on the literature reviewed in Week 6. Letteboer et al. Investigated an interactive multi scale watershed algorithm for segmentation of tumors in brain MRI and compared These results are more comparable with the manually delineated results [1]. Their work demonstrates the use of watershed for analysing tumor structures and boundaries.

Khan et al. proposed a marker-based watershed approach where the contrast enhancement was carried out. before tumor extraction [2]. Padlia and Sharma added gradient information to a marker-controlled In [3] watersheds transform into an improvement on the tumor boundary extraction process. Liu et al. used morphological reconstruction Recently, gradient modification is introduced to suppress unwanted local minima and then perform watershed segmentation [4].

Hafeez et al. studied a morphological and marker based watershed approach that involved the preprocessing stage. and thresholding [5]. Ghaderi et al. investigated the application of a marker controlled watershed along with fuzzy C means. The clustering for glioblastoma segmentation [6] was used.

The above studies show the significance of processing and choice of markers for the extraction of useful watershed regions. According to these results, the proposed implementation involves first performing thresholding and morphological operations to preprocess the image and generate candidate regions, and then performing the watershed transform.

4 Work Completed This Week

4.1 Methodology

The proposed implementation follows a marker-controlled watershed workflow. The main stages are described below.

1. MRI input: A brain MRI slice is selected from the available dataset. The selected image is used as the input for the preprocessing and segmentation stages.

2. Image preprocessing: The input image is converted into a suitable representation for processing. Noise reduction is considered to reduce small intensity variations, while contrast enhancement can improve the visibility of structures in the MRI image.

3. Thresholding: An intensity-based threshold is applied to identify candidate foreground regions. This step provides an initial estimate of the regions that may contain abnormal tissue. Thresholding alone is not treated as the final tumor segmentation.

4. Morphological processing: Morphological operations such as opening and closing are used to remove small unwanted regions and refine the candidate mask. These operations also help prepare the image for marker generation.

5. Marker generation: Foreground and background markers are generated to guide the watershed process. The foreground markers represent regions considered to be inside the objects, while background markers represent regions considered to be outside them. Pixels that cannot be confidently assigned are treated as unknown regions.

6. Watershed segmentation: The marker image and the corresponding three-channel image are supplied to OpenCV's `cv2.watershed()` function. The algorithm expands the labelled regions and identifies boundaries where different regions meet.

7. Segmentation preview: The watershed boundaries are overlaid on the input MRI image for visual inspection. The resulting regions can then be examined to determine whether the candidate tumor area has been separated meaningfully from the surrounding tissue.

4.2 Implementation Details

The implementation is planned around the OpenCV image-processing workflow. The main function used for the watershed stage is `cv2.watershed()`, which accepts a three-channel image and a marker image. The marker image is updated during the watershed process, and boundary pixels are assigned the label `-1`.

The implementation workflow can be summarized as:

1. Read the selected MRI slice.
2. Convert the image into a suitable grayscale representation.
3. Apply noise reduction and contrast enhancement where required.
4. Generate a candidate foreground mask using thresholding.
5. Refine the mask using morphological operations.
6. Calculate distance information to identify reliable foreground regions.
7. Define background and unknown regions.
8. Create the marker image and apply `cv2.watershed()`.
9. Display the watershed boundaries over the original MRI slice.

For OpenCV watershed processing, the input image must have three colour channels. Therefore, a grayscale MRI slice can be converted to a three-channel image before calling the watershed function. The marker image must contain integer labels representing the initial regions.

The distance-transform approach is considered for identifying foreground regions from the refined binary mask. This provides a way to select more reliable internal markers instead of treating every foreground pixel as a separate starting region. The final marker-generation parameters will require testing on the selected MRI images.

5 Results

This week's report presents the implementation workflow and expected output of the marker-controlled watershed method. The intended output is an MRI slice with watershed boundaries displayed over the image, together with the labelled regions generated from the foreground and background markers.

The implementation preview is intended to help inspect whether the generated markers separate the candidate tumor region from surrounding brain tissue. Particular attention will be given to unwanted small regions, incomplete tumor coverage and boundaries that do not follow the visible tumor structure.

No quantitative Dice, IoU, precision or recall values are reported for the watershed method in this report because a completed experimental evaluation has not been established in the information carried forward from Week 6. Once the implementation is tested against the BraTS ground-truth masks, the results can be evaluated using these metrics and compared with the existing threshold-based pipeline.

6 Challenges Faced

The main challenge in watershed segmentation is over-segmentation. Small intensity variations and noise can create multiple local minima, resulting in many unnecessary regions.

Another challenge is the selection of suitable foreground and background markers. If the foreground markers cover only a small part of the tumor, the resulting segmentation may miss tumor regions. If the markers include surrounding tissue, unwanted regions may be included in the output.

Brain MRI images also contain regions with similar intensities. Consequently, thresholding may not always separate tumor tissue from normal brain tissue accurately. The watershed boundary may therefore differ from the expert-annotated tumor boundary.

Parameter selection is another consideration. Threshold levels, morphological kernel size, distance-transform parameters and marker-generation criteria may affect the final segmentation. These parameters need to be examined on multiple MRI images rather than selected from a single example.

7 Plan for Next Week

The next stage will focus on testing and refining the watershed implementation on the selected brain MRI images.

The planned work is:

- Implement and test the complete marker-controlled watershed pipeline using OpenCV.
- Generate and inspect foreground, background and unknown-region markers.
- Test different threshold levels and morphological kernel sizes.
- Visualize the watershed boundaries on the original MRI slices.
- Compare the segmentation output with the available BraTS ground-truth masks.
- Calculate Dice, IoU, precision and recall for the watershed results.
- Compare the watershed output with the existing threshold-based segmentation pipeline.

- Identify improvements needed to reduce over-segmentation and improve tumor boundary accuracy.

References

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